



Ödev - 1

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1- Karmaşıklık Analizi:

Hash:

Input:

t: Int: Key value to be hashed

m: Int: Hash table size

k: Int: Maximum linear probing attempts

table: USER* Array: Hash table array

iterationPtr: Int*: Pointer to store iteration count

user: USER*: User object to be inserted

i=0

firstIndex = t mod m

// Check if first position is empty

IF table[firstIndex] = NULL THEN

 *iterationPtr = 1

 CALL hashTableInsertLinearProbing(user, firstIndex, table)

 RETURN

ENDIF

For i 1 to k-1

 index = (firstIndex + i) mod m

 IF table[index] = NULL THEN

 *iterationPtr = i + 1

 CALL hashTableInsertLinearProbing(user, index, table)

 RETURN

 ENDIF

ENDFOR

// If linear probing fails, use separate chaining

*iterationPtr = k + 1

CALL hashTableInsertSeparateChaining(user, firstIndex, table)

Algoritma da bulunan en karmaşık bölge hash işleminin gerçekleştiği bölgedir. Yukarıda pseudo kodu verilen algoritma linear ve separate chaining birleşimi hash algoritmasını göstermektedir. Bu kodda

for içerisinde kalan işlemler dizi boyutu ile değişmediği için sabit bir sayıdır ve bu kısım 1 olarak değerlendirilmiştir.

$$\sum_{i=1}^{k-1} 1 = k - 1 \equiv \theta(k) \equiv \theta(1)$$

Verilen kod bloğunun karmaşıklığı yukarıdaki denklem ile gösterilmiştir. Bu denklemde iç kısımda kalan bölge açılıp ifade sadeleştirilse verilen kodun ve ödevin karmaşıklığı $\theta(1)$ olarak bulunmaktadır.

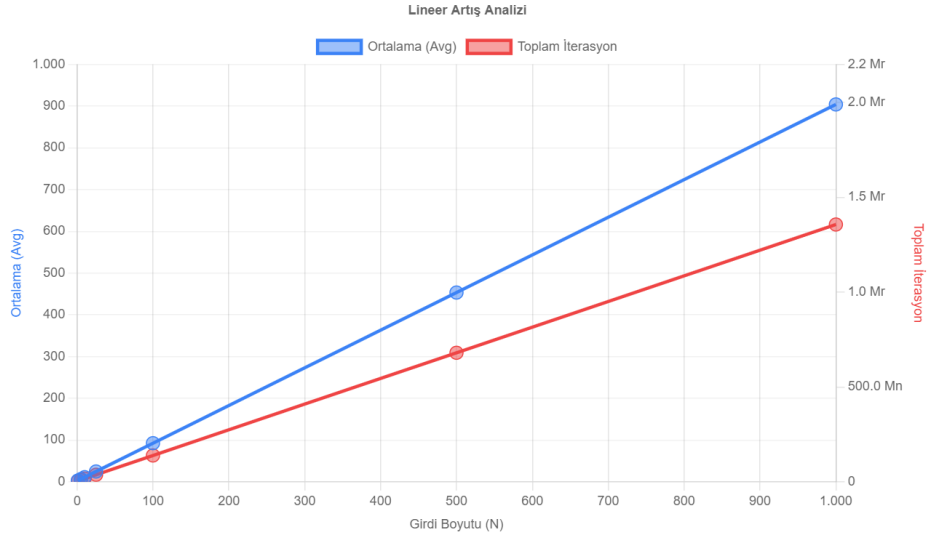
2- Ekran Çıktıları:

- ID-1_5M Input Dosyası Sonuçları

<pre>Lutfen k sayisini girin: 1 Total iterations: 2849999 Average iterations per test: 1.90 ----- Process exited after 1.977 seconds with return value 0 Press any key to continue . . . </pre>	<pre>Lutfen k sayisini girin: 5 Total iterations: 8371698 Average iterations per test: 5.58 ----- Process exited after 2.375 seconds with return value 0 Press any key to continue . . . </pre>
<pre>Lutfen k sayisini girin: 10 Total iterations: 15215501 Average iterations per test: 10.14 ----- Process exited after 2.249 seconds with return value 0 Press any key to continue . . . </pre>	<pre>Lutfen k sayisini girin: 25 Total iterations: 35654990 Average iterations per test: 23.77 ----- Process exited after 5.74 seconds with return value 0 Press any key to continue . . . </pre>
<pre>Lutfen k sayisini girin: 100 Total iterations: 137414833 Average iterations per test: 91.61 ----- Process exited after 3.566 seconds with return value 0 Press any key to continue . . . </pre>	<pre>Lutfen k sayisini girin: 500 Total iterations: 678666318 Average iterations per test: 452.44 ----- Process exited after 18.11 seconds with return value 0 Press any key to continue . . . </pre>
<pre>Lutfen k sayisini girin: 1000 Total iterations: 1354631935 Average iterations per test: 903.09 ----- Process exited after 5.214 seconds with return value 0 Press any key to continue . . . </pre>	

Performans Analiz Grafiği

Girdi boyutuna göre Toplam İterasyon ve Ortalama değerlerin değişimi



- ID-2 Input Dosyası Sonuçları(2000000 unique ID)

<pre>Lutfen k sayisini girin: 1 Total iterations: 3800001 Average iterations per test: 1.90 ----- Process exited after 2.385 seconds with return value 0 Press any key to continue . . .</pre>	<pre>Lutfen k sayisini girin: 5 Total iterations: 11061486 Average iterations per test: 5.53 ----- Process exited after 3.476 seconds with return value 0 Press any key to continue . . .</pre>
<pre>Lutfen k sayisini girin: 10 Total iterations: 20109912 Average iterations per test: 10.05 ----- Process exited after 2.465 seconds with return value 0 Press any key to continue . . .</pre>	<pre>Lutfen k sayisini girin: 25 Total iterations: 47181283 Average iterations per test: 23.59 ----- Process exited after 2.475 seconds with return value 0 Press any key to continue . . .</pre>
<pre>Lutfen k sayisini girin: 100 Total iterations: 182373520 Average iterations per test: 91.19 ----- Process exited after 2.951 seconds with return value 0 Press any key to continue . . .</pre>	<pre>Lutfen k sayisini girin: 500 Total iterations: 902842302 Average iterations per test: 451.42 ----- Process exited after 5.183 seconds with return value 0 Press any key to continue . . .</pre>
<pre>Lutfen k sayisini girin: 1000 Total iterations: 1803232399 Average iterations per test: 901.62 ----- Process exited after 6.496 seconds with return value 0 Press any key to continue . . .</pre>	

Performans Analiz Grafiği

Girdi boyutuna göre Toplam İterasyon ve Ortalama değerlerin değişimi

